BACTERIA STORM SIMULATION

The 'critical bacteria storm' was developed to address bacteria impairments in Los Angeles County coastal watersheds for protection of recreational water quality (LARWQCB 2018). The critical bacteria storm is the depth of precipitation associated with the 90th percentile wet day when bacteria receiving water limitations (RWLs) apply. For water quality planning purposes, bacteria RWLs are assumed to *not* apply across waterbodies during either of the following conditions:

- **Allowable exceedance days** ensure that stormwater compliance plans are not required to achieve bacteriological water quality that is higher than undeveloped reference watersheds. The quantity and applicability of allowable exceedance days are determined by regional/waterbody specific Total Maximum Daily Loads (TMDLs; LARWQCB 2019).
- **High flow suspension (HFS) days** are those with greater than or equal to 0.5 inches of rainfall plus the following 24-hour period. The applicability of HFS days is prescribed by the Basin Plan for the Coastal Watersheds of Los Angeles and Ventura Counties (LARWQCB 2014). For illustration, a high flow suspension day for the Los Angeles River is shown in Figure 1 below.

Figure 1. High flow suspension day along Los Angeles River Reach 3 at the Glendale Narrows.

To facilitate the simulation of critical bacteria storms with WMMS 2.0, the WMMS 2.0 Repository has made available a number of "Bacteria Storm Input File Groups" for download (Table 1). Each grouping contains the 906 precipitation files to simulate precipitation and runoff across the LSPC model are. Reference the *WMMS 2.0 Phase I Report: Model Calibration and Validation* as needed (also available in the WMMS 2.0 Repository). Note, the number of applicability and quantity of Allowable Exceedance Days and HFS Days should be evaluated prior to any simulation of the critical bacteria storm. The Bacteria Storm Input File Groups below are provided for convenience only and their applicability to any waterbody should be confirmed before use.

Table 1. Waterbodies, associated conditions, and WMMS 2.0 Bacteria Storm Input File Group.

Waterbody/Bacteria TMDL	Number of Allowable Exceedance Days[1]	HFS Applicable?	Bacteria Storm Input File Group
San Gabriel River (Non-HFS waterbodies)	17	No	1
Malibu Creek (below Pacific Coast Highway)	17	No	
Marina Del Rey	17	No	
Santa Monica Bay Beaches	17	No	
Santa Clara River	16	No	2
Los Angeles River (Non-HFS waterbodies)	15	No	3
Malibu Creek (above Pacific Coast Highway)	15	No	
Ballona Creek[2]	15	Yes	4
San Gabriel River (HFS waterbodies) [3]	11	Yes	5
Los Angeles River (HFS waterbodies) [3]	10	Yes	6

[1] Based on daily sampling
[2] When the number of HFS days is <= 15, RWLs apply on the 16th wet day. If the number of HFS days is > 15, RWLs apply on the first non-HFS wet day (for additional detail see the example on the next page).
[3] Allowable Exceedance Days apply after all the HFS day have been applied (regardless of how many HFS days occur)

Temporal distribution of the critical bacteria storm depth in each precipitation file is based upon the LACDPW 4-day unit hyetograph for typical storm banding (LACDPW 2006), as shown in Figure 2. This unit hyetograph was scaled to the corresponding cumulative precipitation depth for each subwatershed.

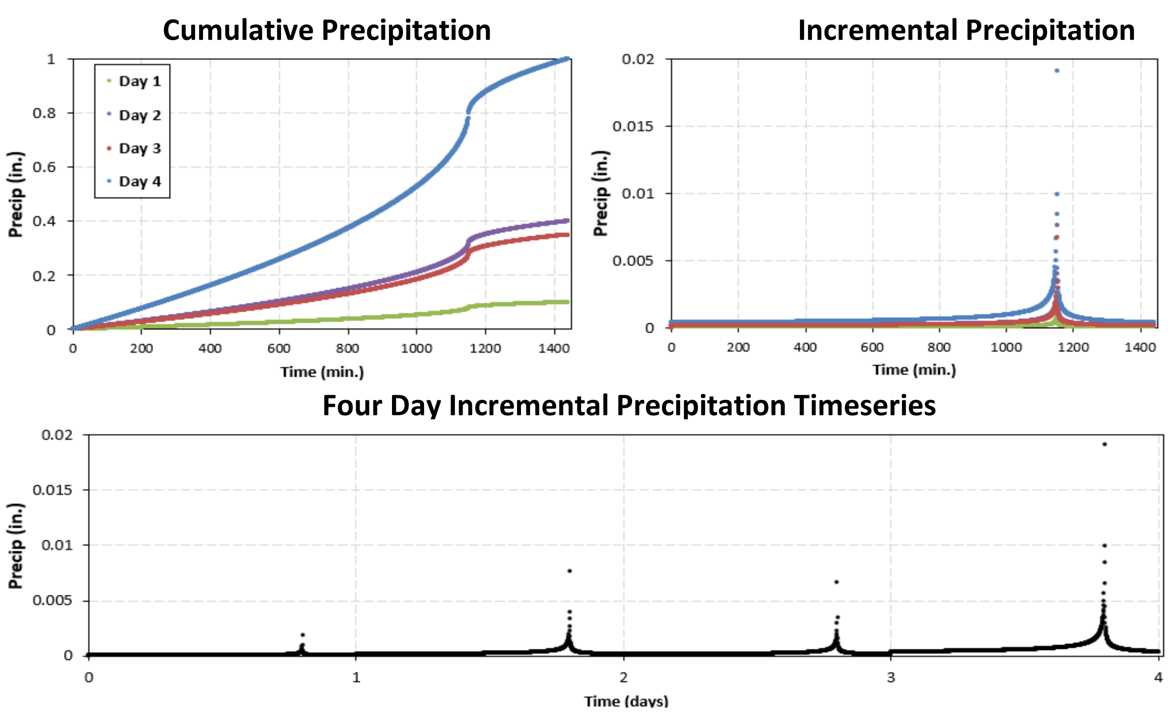

Figure 2. Cumulative (top left) and incremental precipitation (top right and bottom) for the 1-inch unit storm using the LACDPW hyetograph and 4-day design storm (LACDPW 2006).

On the next page, an example calculation of the bacteria storm depth calculation for LACDPW hourly ALERT Gage 482 (USC) is provided. This gage is in Ballona Creek watershed; therefore, it is in Bacteria Storm Input File Group 4 where HFS is applicable and up to 15 exceedance days can be applied.

Using the Ballona Creek TMDL, Table 2 lists all of the **HFS days** for each year followed by wet days in descending order, up to the 21st day, for the most recent 10-years of data. Wet days are those having 0.1 inches of precipitation or greater plus the following 3 days. If the quantity of HFS days for a year is greater than or equal to 15, the first day RWLs apply is the first non-HFS wet day. If the quantity of HFS days for a year is less than 15, the first day RWLs apply is 16th wet day (including the count of all HFS days). In the table below, the first day RWLs apply is marked as a red, dashed box. The 90th percentile value for these days (the second highest value across the 10-year period) is 0.47 inches.

Table 2. Calculation of critical bacteria storm depth for Ballona Creek TMDL for the most recent 10 water years with hourly ALERT Gage 482 (USC) in Ballona.

Day	2009		2010		2011		2012		2013		2014[1]		2015		2016		2017		2018[1]	
	Date	In.	Date	In.	Date	In.	Date	In.	Date	In.	Date	In.	Date	In.	Date	In.	Date	In.	Date	In.
1	12/15	1.34	2/6	1.57	3/20	2.84	10/5	0.94	1/24	0.63	2/28	1.81	9/15	2.14	1/5	1.4	1/22	1.76	1/9	1.48
2	11/26	1.26	10/14	1.14	12/19	2.53	3/25	0.79	5/6	0.63	2/27	0.86	12/12	1.48	1/6	0.68	2/17	1.67	3/21	0.62
3	2/16	0.67	1/20	0.95	12/22	2.44	11/20	0.78	3/8	0.51	3/1	0.83	12/2	0.88	2/17	0.56	1/20	1.38	3/22	0.5
4	2/5	0.59	1/18	0.90	12/20	1.4	3/17	0.63	1/25	0.11	3/2	0.2	5/14	0.73	3/6	0.55	12/23	1.15	1/10	0.01
5	2/6	0.55	1/21	0.87	12/18	1.18	4/11	0.59	3/9	0	11/21	0.24	2/22	0.62	3/11	0.53	12/16	1.06	3/23	0
6	2/17	0.31	12/7	0.79	12/21	0.96	12/12	0.59	5/7	0	4/2	0.19	12/3	0.25	3/7	0.35	1/12	0.93	3/10	0.4
7	12/16	0.04	12/12	0.75	12/29	0.71	1/21	0.55	12/18	0.43	11/29	0.16	5/15	0.03	1/7	0.26	1/19	0.77	3/2	0.31
8	11/27	0	4/12	0.63	3/21	0.59	12/13	0.12	12/29	0.4	2/2	0.12	2/23	0.01	2/18	0.17	2/6	0.68	1/8	0.27
9	2/7	0	2/27	0.63	2/19	0.59	3/18	0.08	11/30	0.36	11/22	0.03	12/13	0	3/12	0	1/9	0.63	3/15	0.15
10	12/17	0.47	4/5	0.55	2/25	0.56	11/21	0.04	12/24	0.32	11/23	0	9/16	0	1/31	0.36	1/23	0.43	3/16	0.1
11	2/9	0.43	2/5	0.51	2/16	0.52	3/26	0.04	12/26	0.31	11/24	0	1/11	0.47	10/5	0.34	2/7	0.19	3/3	0.04
12	2/13	0.28	1/19	0.47	2/26	0.45	10/6	0	11/17	0.24	11/30	0	3/1	0.46	12/19	0.26	12/24	0.17	3/11	0.03
13	11/25	0.24	1/22	0.35	2/20	0.05	1/22	0	2/19	0.16	12/1	0	1/10	0.41	12/13	0.14	1/10	0.05	3/13	0.02
14	3/4	0.20	12/13	0.28	12/23	0	4/12	0	11/29	0.15	12/2	0	12/16	0.38	4/8	0.13	2/18	0.05	1/11	0
15	12/25	0.16	10/15	0	12/30	0	1/23	0.47	12/3	0.15	2/3	0	11/1	0.38	12/22	0.06	12/17	0	1/12	0
16	1/23	0.16	12/8	0	2/17	0	4/13	0.43	1/6	0.12	2/4	0	3/2	0.37	4/9	0.03	1/13	0	3/4	0
17	6/5	0.12	2/7	0	3/22	0	11/6	0.32	12/1	0.04	2/5	0	7/18	0.34	3/14	0.02	1/21	0	3/5	0
18	11/4	0.12	2/28	0	12/25	0.49	4/26	0.27	12/2	0.04	3/3	0	11/30	0.26	12/21	0.01	12/21	0.47	3/12	0
19	1/24	0.08	4/6	0	2/18	0.48	11/4	0.16	1/26	0.04	3/4	0	5/8	0.16	10/6	0	11/20	0.43	3/17	0
20	11/5	0	4/13	0	1/2	0.47	4/25	0.16	1/27	0.04	3/5	0	12/17	0.13	10/7	0	1/11	0.39	3/18	0
21	11/6	0	12/11	0.47	3/23	0.46	11/12	0.12	11/18	0	4/3	0	12/30	0.13	10/8	0	2/10	0.3	3/19	0
# of HFS days in record		9		20		17		14		6		4		10		9		17		5

The 90th percentile depth identified in Table 2 (0.47 in) and associated 4-day timeseries is only applicable to the subwatersheds within the precipitation zone it encompasses. Figure 3 illustrates this precipitation zone and subwatersheds for this timeseries (white dashed line). It also shows the depths for the other zones in the Ballona Creek Watershed Management Group area. Reference the *LSPC Boundary Conditions Reference Shapefiles* on the WMMS 2.0 Repository to "crosswalk" precipitation zones, weather stations, and subwatersheds.

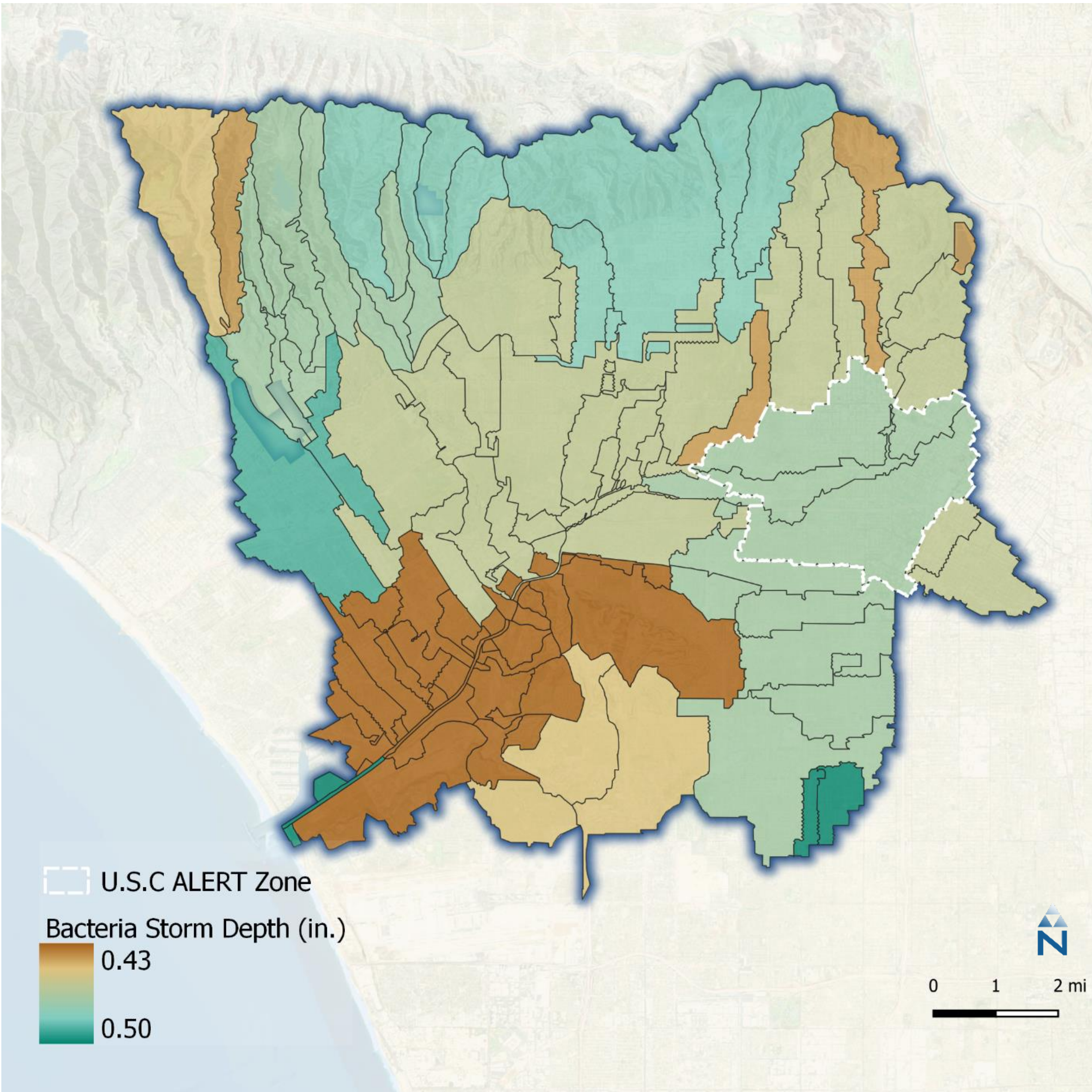

Figure 3. Precipitation zone depths across Ballona Creek. The hourly ALERT Gage 482 (USC) zone is highlighted.

REFERENCES

LACDPW (Los Angeles County Department of Public Works). 2006. Hydrology Manual. Water Resources Division. January 2006. Retrieved January 2020 from https://dpw.lacounty.gov/wrd/publication/engineering/2006_Hydrology_ Manual.pdf

LARWQCB (Los Angeles Regional Water Quality Control Board). 2014. Basin Plan for the Coastal Watersheds of Los Angeles and Ventura Counties. Sept. 11, 2014. Retrieved January 2020 from https://www.waterboards.ca.gov/losangeles/water_issues/programs/basin_plan/basin_plan_documentation.html

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